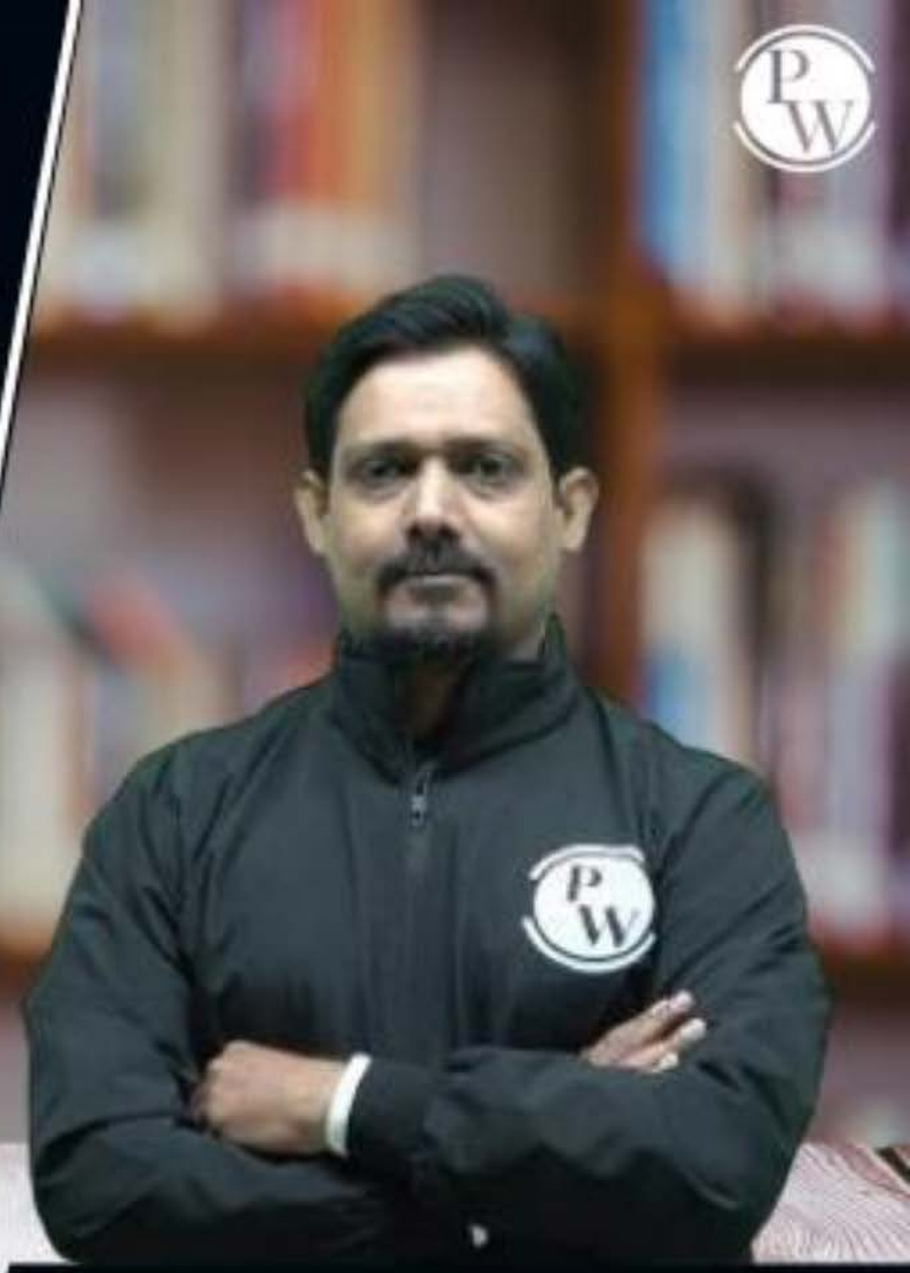
Computer Science & IT



Data Structure & Programming

Stack

Lecture No. 01



By- Abhishek Sir

Recap of Previous Lecture



Topic — 3-D array

Topic — Tridiagonal Matrix

Topic — Complexity

Topic

Topic

Slide

Topics to be Covered



- Topic — Stack
- Topic — Implementation of stack
- Topic
- Topic
- Topic

Slide



Topic : Stack as LIFO

1. Stack is a Linear data.
2. Stack is one ended data structure.
3. Stack is Last in first out (LIFO).
4. Operation perform on on end of collection of element indicated by top

Slide



Topic : Stack as LIFO

5. Insert of an element is called push
   Deletion of element is called pop
   } top of stack

6. Application
   - Recursion
   - permutation generator
   - Expression in Computer : Infix expression
     prefix expression
     postfix expression

Slide



Stack



top = top + 1;

top = -1



Stack



top = top+1 (0)

Insert(2)

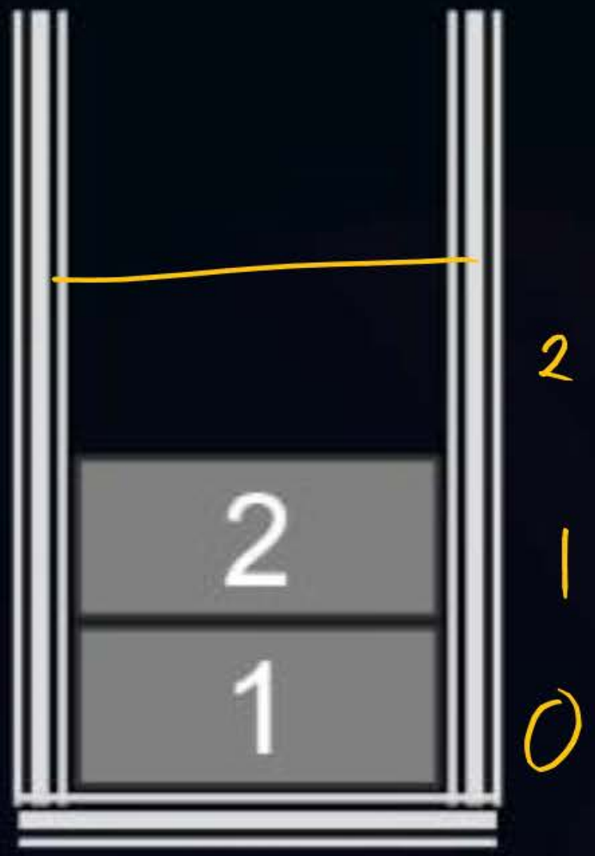
a[top] = data

top = 0 ✓

top = top+1 . 1



Stack



top = 1

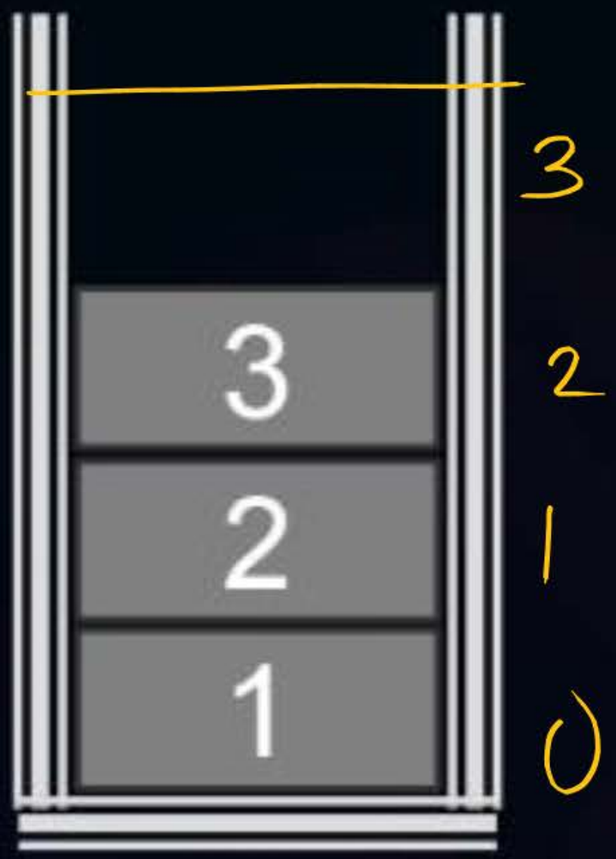
Insert(3)

top = 1+1 = 2

a[top] = 3



Stack



top = 2

Insert(4)

top = top + 1 ;

2 + 1 = 3

a[top] = 4



Stack



top=3

pop()

data 4

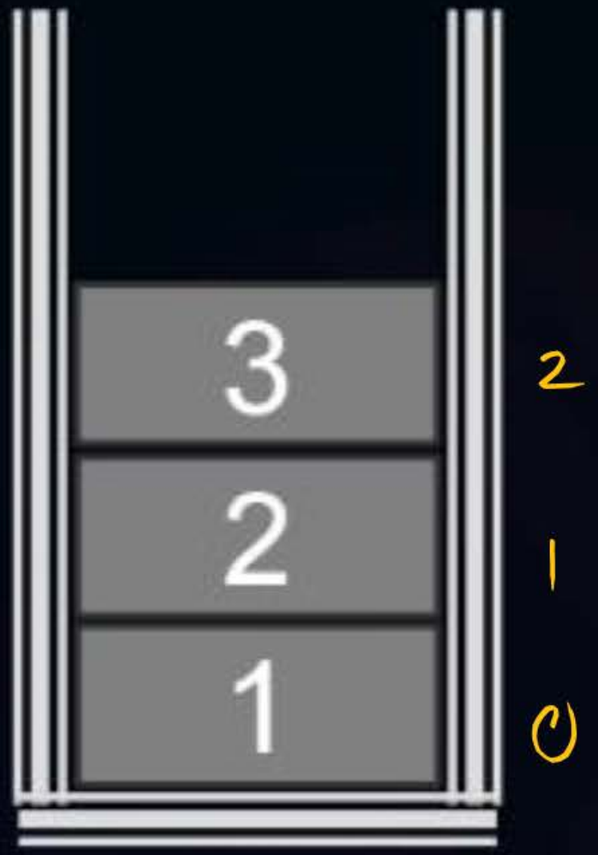
data = a[top]

data = a[3]

top: top-1



Question



top = ~~2~~ 1

pop

data : 3

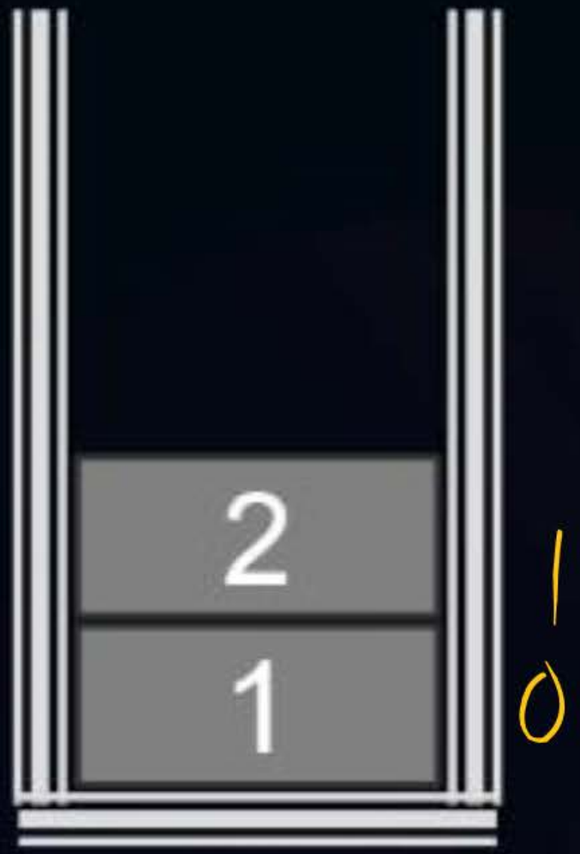
data = a[top];

data = a[2] = 3

top = top - 1



Stack



top: ~~1~~ 0

pop()

data = 2

data = a[top]

= a[1]

data = 2

top: top - 1



Stack



top = ∅ - 1

data = a[0]
= 1

top = top - 1

= -1



Stack



An empty stack



Question

#Q The following sequence of operations is performed on a stack:

PUSH (10), PUSH (20), POP , PUSH (10), PUSH (20), POP, POP, POP, PUSH (20), POP

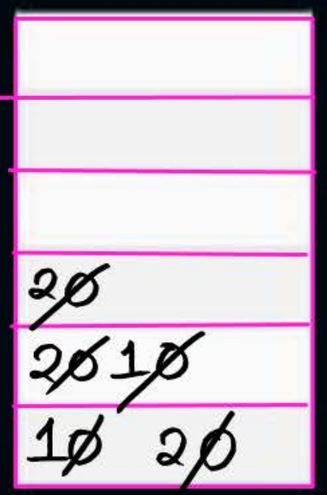
The sequence of values popped out is:

(A) 20, 10, 20, 10, 20

(B) 20, 20, 10, 10, 20

(C) 10, 20, 20, 10, 20

(D) 20, 20, 10, 20, 10





Question

#Q Consider the usual algorithm for determining whether a sequence of parentheses is balanced. What is the maximum number of parentheses that will appear on the stack AT ANY ONE TIME when the algorithm analyzes: (()(())(()))?

(A) 1

(B) 2

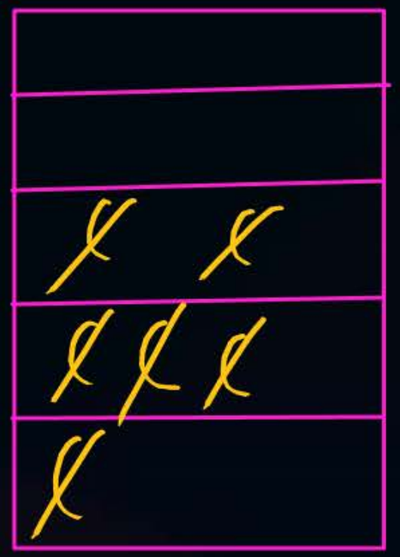
(C) 3

(D) 4

(- open
push

) - closing

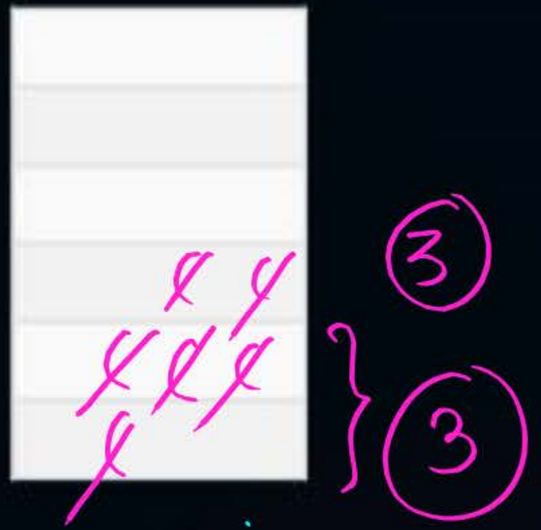
} stack





Question

#Q Consider the usual algorithm for determining whether a sequence of parentheses is balanced. What is the maximum number of parentheses that will appear on the stack AT ANY ONE TIME when the algorithm analyzes: (()(())(()))?



(A) 1

(B) 2

(C) 3

(D) 4



Topic : Stack Implementation

Array Implementation

```
#define Max 100
```

← Macro — preprocessor

```
int a[Max];
int top;
```

} Global

Slide



Topic : Stack Implementation

void init ( )	int isEmpty()	void push(int)	int pop()
Initialize the top top - 1	return 1 (true) if stack is empty return 0 if Stack is Nonempty	Insert the element at top of stack	Delete the element from top of stack

Slide



Topic : Stack Implementation

```
Void init () {
    top = -1;
}
```

```
int isEmpty ( ) {
    if (top == -1)
        return 1;
    else
        return 0;
}
```

```
int isEmpty ( ) {
    return (top == -1);
}
```

Slide



Topic : Stack Implementation

```
void push(int data){
    if(top == Max-1){
        printf("stack is full");
        return;
    }
    top = top+1;
    a[top] = data;
}
```

parameter

Increment the top Insert the value

```
a[++top] = data
```

modifyable

Slide



Topic : Stack Implementation

```
int pop() {
    int data;
    if (top == -1) {
        printf("empty stack");
        return -1
    }
    data = a[top];
    top = top - 1;
    return data;
}
```

if (isEmpty())

stack underflow

Slide



Topic : Stack Implementation

Question: In above implementation of stack, following operation are performed.

I. push 1 ... 50 element In reverse sequence ← 50

II. pop 10 element from stack ← 10 No. of element = 40

(III) Insert 30 Number. 51 ... 80 in given order

30 Insert

What is the value of top ___

70

Slide



Topic : Stack Implementation

Index top starts from 0 — 69

value of top = 69

value at the top = 80

Slide



Topic : Stack Implementation

Question: In above implementation of stack, following operation are performed.

I. push 1 ... 50 element In reverse sequence

II. pop 10 element from stack

(III) Insert 30 Number . 51 ... 80 in given order

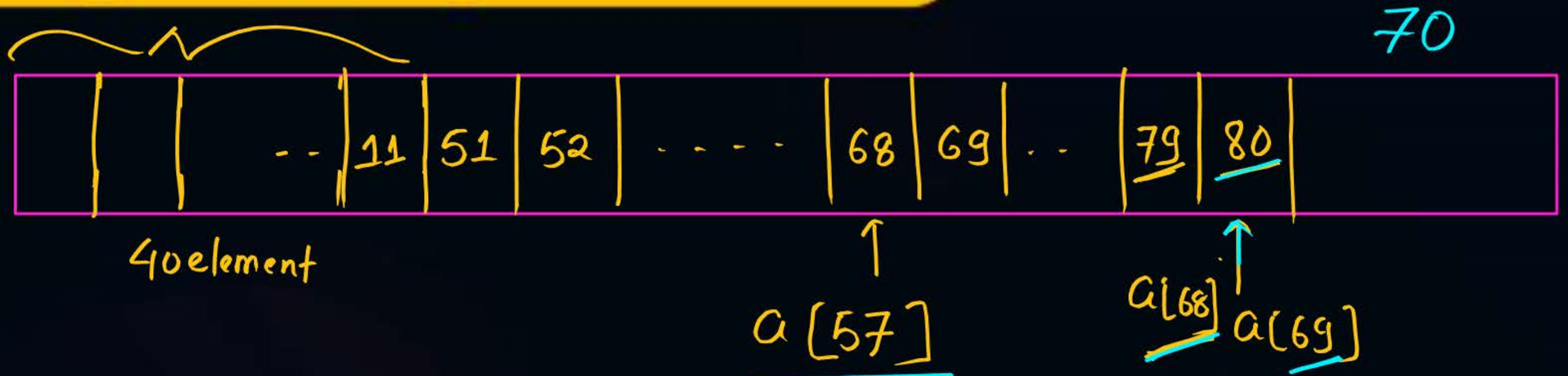
Suppose the BA of array is 1000 the at what is the Address of element 68

Size = 2B

Slide



Topic : Stack Implementation



$1000 + 57 \times 2$

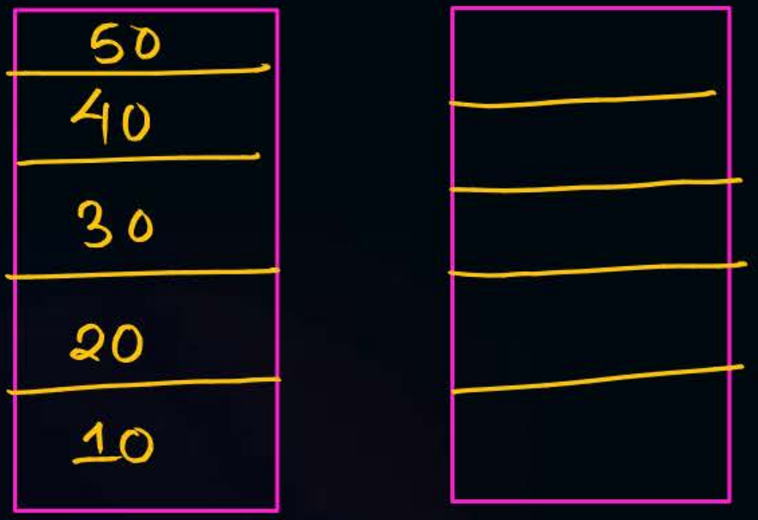
$1000 + 114 = 1114$

$BA + (i - 0) \times s$

Slide

Topic : Stack Implementation

Question



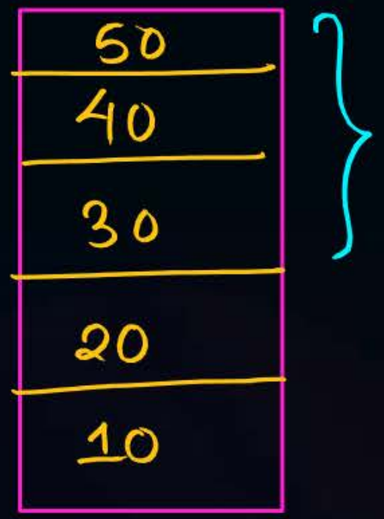
S_1 S_2

Consider given two stack S_1 and S_2. S_1 consists of 5 element 10,20,30,40,50 and S_2 is empty.

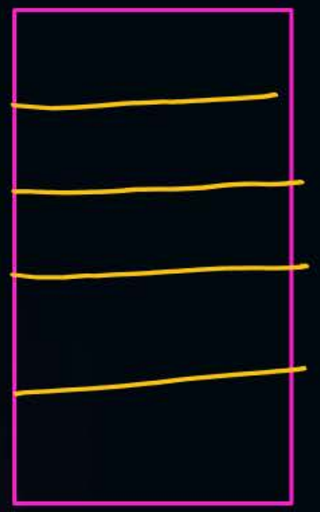
We Need to pop 10 & 20 the remaining will be present in stack is same order i.e. 50 on top of stack. We can use S_2 as temporary storag the minimum push & pop(total) for performing the operation

Topic : Stack Implementation

Question



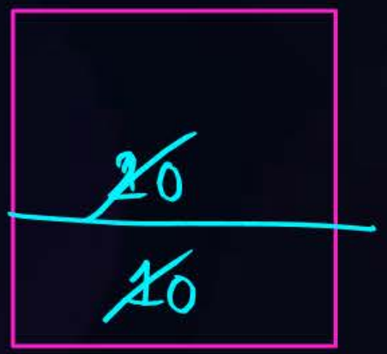
S_1

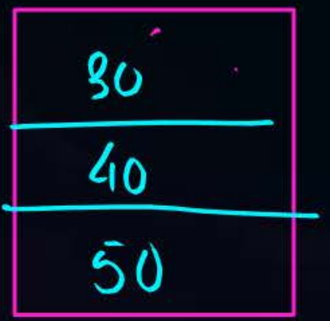


S_2



S_1





$x = pop()$ ✓

$push(x) - S_2$ ✓

$x = pop() \; S_1$ ✓

$push(x) \; S_2$ ✓

$x = pop \; S_1$ ✓

$push(x)$ ✓

$pop \; S_1$

$pop \; S_1$

$x = pop \; S_2$

$push(x) \; S_1$

$x = pop() \; S_2$

$push(x) \; S_1$

$x = pop \; S_2$

$push(x) \; S_1$

14



2 mins Summary

Topic Stack as Linear Data structure

Topic Stack Implementation

Topic push , pop , init , isEmpty

Topic

Topic

Slide

THANK - YOU